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months. Typically, interest rates are quoted in annual terms, so the interest rate on a 90-day government debt security quoted at 3 percent is the annualized rate and not the actual interest rate earned over the 90-day period.

Whether the interest rate we use is a required rate of return, or a discount rate, or an opportunity cost, the rate encompasses the real risk-free rate and a set of risk premia that depend on the characteristics of the cash flows. The foundational set of premia consist of inflation, default risk, liquidity risk, and maturity risk. All these premia vary over time and continuously change, as does the real risk-free rate. Consequently, all interest rates fluctuate, but how much they change depends on various economic fundamentals—and on the expectation of how these various economic fundamentals can change in the future.

EXAMPLE 1

Determining Interest Rates

Exhibit 1 presents selected information for five debt securities. All five investments promise only a single payment at maturity. Assume that premiums relating to inflation, liquidity, and default risk are constant across all time horizons.

Exhibit 1: Investments Alternatives and Their Characteristics

Investment	Maturity (in years)	Liquidity	Default Risk	Interest Rate (%)
1	2	High	Low	2.0
2	2	Low	Low	2.5
3	7	Low	Low	r_3
4	8	High	Low	4.0
5	8	Low	High	6.5

Based on the information in Exhibit 1, address the following:

1. Explain the difference between the interest rates offered by Investment 1 and Investment 2.

   Solution:

   Investment 2 is identical to Investment 1 except that Investment 2 has low liquidity. The difference between the interest rate on Investment 2 and Investment 1 is 0.5 percent. This difference in the two interest rates represents a liquidity premium, which represents compensation for the lower liquidity of Investment 2 (the risk of loss relative to an investment's fair value if the investment needs to be converted to cash quickly).

2. Estimate the default risk premium affecting all securities.

   Solution:

   To estimate the default risk premium, identify two investments that have the same maturity but different levels of default risk. Investments 4 and 5 both have a maturity of eight years but different levels of default risk. Investment 5, however, has low liquidity and thus bears a liquidity premium relative to Investment 4. From Part A, we know the liquidity premium is 0.5 percent. The difference between the interest rates offered by Investments 5 and 4 is 2.5 percent (6.5% – 4.0%), of which 0.5 percent is a liquidity premium. This